

$$\kappa=\frac{w''}{[1+(w')^2]^{3/2}}$$

$$\frac{3}{2}(w')^2w''$$

$$F_k(x)=k_1x+k_3x^3, k_3>0$$

$$V(x)=\int_0^x\left(k_1\xi+k_3\xi^3\right)d\xi=\frac{1}{2}k_1x^2+\frac{1}{4}k_3x^4, k_3>0$$

$$k_3<0$$

$$F_k(x)=k_1x+k_3x^3, k_3<0$$

$$\frac{|k_3|}{c_2|\dot{x}|^2}F_d=\frac{F_d}{\mu N(\dot{x})}=\frac{F_d}{x}=\frac{F_k(x)}{x}$$

$$k_3>0, k_3<0, k_3=0, x_1(t), x_2(t), \alpha x_1+\beta x_2, \mathcal{N}(\alpha x_1+\beta x_2)\neq \alpha \mathcal{N}(x_1)+\beta \mathcal{N}(x_2), \ddot{x}+x+\varepsilon x^3=0$$

$$\begin{array}{c} x_1\\ x_2\\ \ldots\\ (x_1+x_2)+\\ (x_1+\\ x_2)+\\ \varepsilon(x_1+\\ x_2)^3=\\ \varepsilon(3x_1^2x_2+\\ 3x_1x_2^2)\neq\\ 0\\ k_{eq}\\ A\\ \omega_n(A)\approx\omega_0\left(1+\frac{3}{8}\varepsilon A^2\right)\left(Duffing\right) \end{array}$$

$$\begin{array}{c} \omega_0=\\ \sqrt{k_1/m}\\ \varepsilon>0\\ \varepsilon<0\\ \omega/\omega_n\\ A\\ ,\\ *)^2+\\ 4*\\ 0.05*\\ 0.05*\\ \mathfrak{A}\mathfrak{B}+0.0001*(\mathfrak{A}\mathfrak{B}0.8?(\mathfrak{A}0.8)^3:0));[red,thick,-Stealth](3.8,3.0)-\\ -(5.5,3.0);[red,thick,-Stealth](5.8,0.9)-\\ -(4.2,0.9);[red]at(7.0,2.6);[red]at(3.0,1.2);[gray,thin,dashed](3.5,0)-\\ -(3.5,4.5);[gray,thin,dashed](5.5,0)-\\ -(5.5,4.5);at(3.5,-0.4)\omega_1\\ \omega_2\\ 2+\\ 4*\\ 0.05*\\ 0.05*\\ *)\big)\big);[green!50!black]at(7.5,1.0);\\ \omega_1\\ \omega_2^2\\ \omega_1^2\\ \omega_1<\\ \omega<\\ \omega_2\\ \omega_n\\ \text{Superharmonic} \end{array}$$